

Beyond "Being There": An Integrated Framework for Measuring Narrative Experience in Immersive Virtual Reality

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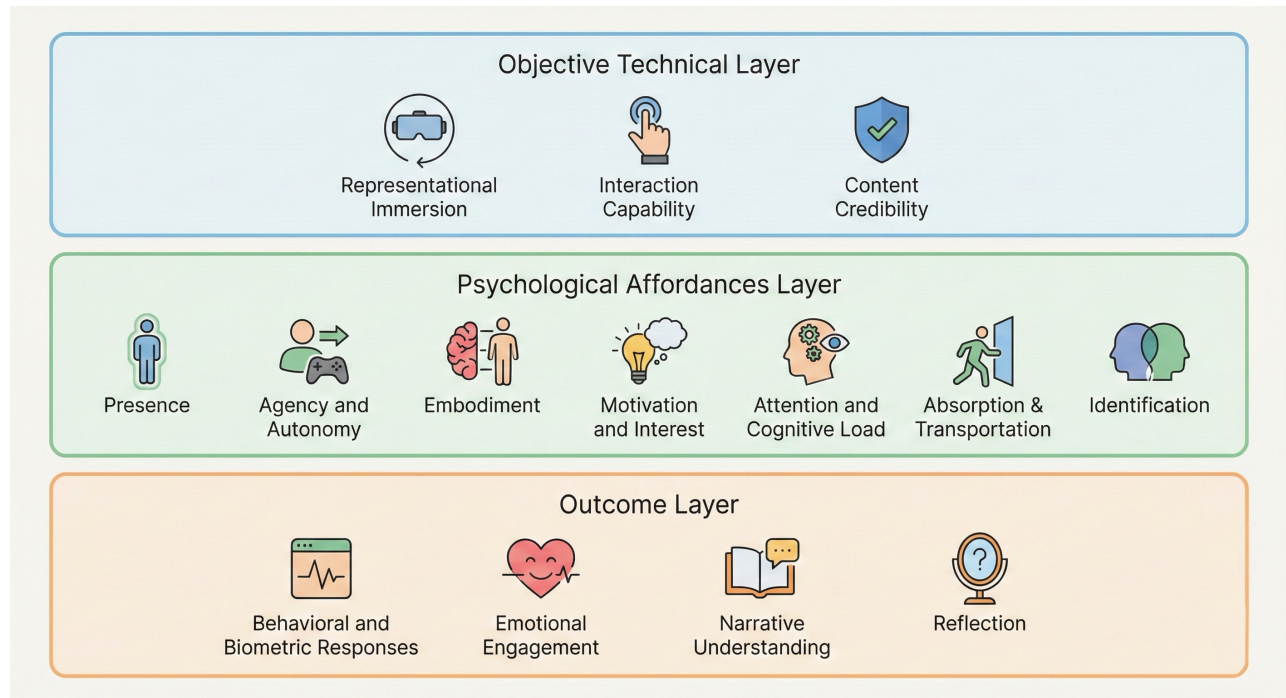


Figure 1: The proposed multi-dimensional framework for VR narrative experiences.

Abstract

Virtual Reality (VR) holds transformative potential for narrative experiences through its superior sensory fidelity and interactivity. However, the strategic integration of narratology and VR technology remains in its infancy. While extensive research addresses "immersion" and "narrative" as independent constructs, empirical studies specifically focusing on their intersection—VR Narrative—are notably scarce. This paper systematically reviews existing literature to identify and summarize the experience dimensions that drive immersive storytelling. By adapting and refining the Cognitive Affective Model of Immersive Learning's framework, we propose a framework to classify, integrate, and map current VR narrative assessment dimensions and questionnaires. Based on these

insights, we identify under-researched gaps and propose future directions. This work provides a clear roadmap for establishing a robust evaluation system for deeply immersive and interactive narrative experiences.

CCS Concepts

• **Human-centered computing** → HCI theory, concepts and models; *User studies; Virtual reality.*

Keywords

Virtual Reality (VR), Narrative Experience, Transportation, Questionnaire



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1 Introduction

Virtual Reality (VR) represents the new frontier of immersive storytelling. By leveraging the pivotal mechanisms of immersion and interactivity, VR facilitates a fundamental identity shift from "spectator" to "participant" [18]. However, as Makransky [18] posits, the VR's leap is not a byproduct of the medium itself, but rather the result of a deep integration between design and VR's unique affordances.

Despite VR's rapid proliferation, current User Experience (UX) measurement paradigms remain disproportionately focused on spatial and technological factors. Established instruments, such as the Presence Questionnaire (PQ) [36] and the Slater, Usoh, and Steed (SUS) scale [33], predominantly assess Spatial Presence and sensory fidelity. While these metrics effectively measure the "Place Illusion", they often fail to capture the nuances of the "Plausibility Illusion" [28]. Consequently, critical psychological states associated with deep narrative processing, such as Narrative Transportation [9] and Narrative Engagement [4], remain underrepresented in empirical research [5].

As hardware becomes more mature, factors like visual coherence, aesthetics, and emotional resonance are beginning to supersede the influence of Presence alone in determining user satisfaction [5]. Furthermore, Green & Brock [9] argue that potent narrative transportation can suppress a user's capacity for counter-arguing, implying that superior narrative design can offset minor technical imperfections, such as latency or resolution issues, thereby sustaining immersion through the Plausibility Illusion [28]. Moving from "shallow sensory stimulation" to "deep narrative engagement" requires a multidimensional evaluation framework that explicitly defines how cognition and emotion flow within virtual environments.

This paper systematically reviews VR narrative dimensions and assessment tools to bridge the gap between technical immersion and narrative depth. Our contributions include:

- Harmonizing VR narrative dimensions across immersive contexts.
- Categorizing existing questionnaires by their corresponding narrative dimensions.
- Integrating isolated constructs from these disparate fields, provides a unified language for both narrative designers and technical researchers.
- Identifying high-potential, underexplored evaluation metrics for future study.

2 Literature Review

In the fields of Film Studies and Narratology, established scales exist to measure constructs like emotional involvement, mental imagery and character identification [1, 9, 12, 15]. However, these tools are primarily designed for non-interactive, linear media (text, video and weakly interactive 360-degree VR) [1, 9], as well as generalized cross-media scales [4, 15]. Virtual Reality (VR) transforms the audience's role from a "witness" to an "embodied participant" within the environment. These scales do not account for the unique characteristics of immersive VR, such as presence, agency and interactivity [18, 28], which can cause traditional metrics to lose validity in VR

contexts. Consequently, precise measurement standards for core VR narrative dimensions remain limited.

Although early assessment tools like the Presence Questionnaire (PQ) [36] and the SUS Scale [33] primarily focused on presence and technical parameters, recent research has shifted toward systematic integration frameworks. Hameed et al. [10] proposed a multidimensional quality taxonomy to disentangle the interrelated factors shaping VR experiences and support systematic evaluation. However, it lacks a quantitative evaluation for psychological differences driven by deep narrative logic or plot. Similarly, the UXIVE model developed by Tcha-Tokey et al. [31] covers ten components, including flow and emotion, yet it was designed for "edutainment" tasks, emphasising efficiency and usability over the unique aesthetic resonance of narrative works. Furthermore, while the The Cognitive Affective Model of Immersive Learning (CAMIL) [18] framework provides a structured perspective for immersive learning, its goal is knowledge acquisition rather than narrative transportation, thereby overlooking the deep emotional resonance generated by story structures. In summary, while existing frameworks have improved evaluation holistically, a significant gap remains in the systematic quantification of narrative-driven experiences.

3 VR Narrative Experience Framework

To systematically deconstruct the complexity of the VR narrative experience, our framework functional synthesis that re-purposes CAMIL [18] for the narrative domain. While the original CAMIL focuses on knowledge acquisition (learning), we pivot the core objective to Narrative Transportation (being absorbed into the story).

We bridge Slater's immersion theory with CAMIL by identifying Place Illusion (PI) [28] and Plausibility Illusion (Psi) as one of the primary catalysts for Psychological Affordances, and innovatively integrating dimensions such as Narrative Transportation [9, 12] to establish this hierarchical (L1-L2-L3) evaluation framework. The framework establishes a clear causal trajectory from "objective stimuli" to "subjective illusions" and ultimately to "behavioral responses". This structure effectively addresses the critical disconnect between technical metrics and narrative experience prevalent in existing evaluation paradigms.

Objective Technical Layer (Level 1) focuses on the physical attributes of the narrative medium—defined as immersion—which serves as the essential technological foundation for inducing psychological illusions [3, 28, 36]. This layer is primarily driven by *Representational Immersion*, where sensory fidelity factors such as resolution, field of view (FOV), and sensory channel richness simulate the perceptual laws of the physical world to generate the "Place Illusion" (PI) [3, 28, 36]. Parallel to sensory input, *Interaction Capability* utilizes tracking precision and immediate feedback to establish a tight causal link between user actions and environmental responses, thereby evoking the "Plausibility Illusion" (Psi) [2, 3, 7, 8, 20, 21, 28]. To sustain this psychological state, *Content Credibility* integrates narrative logic, plot consistency, and world coherence to ensure that the virtual environment remains believable and prevents the "breakdown of the illusion" [4, 29]. As detailed in Fig. 2, these objective dimensions are mapped to established

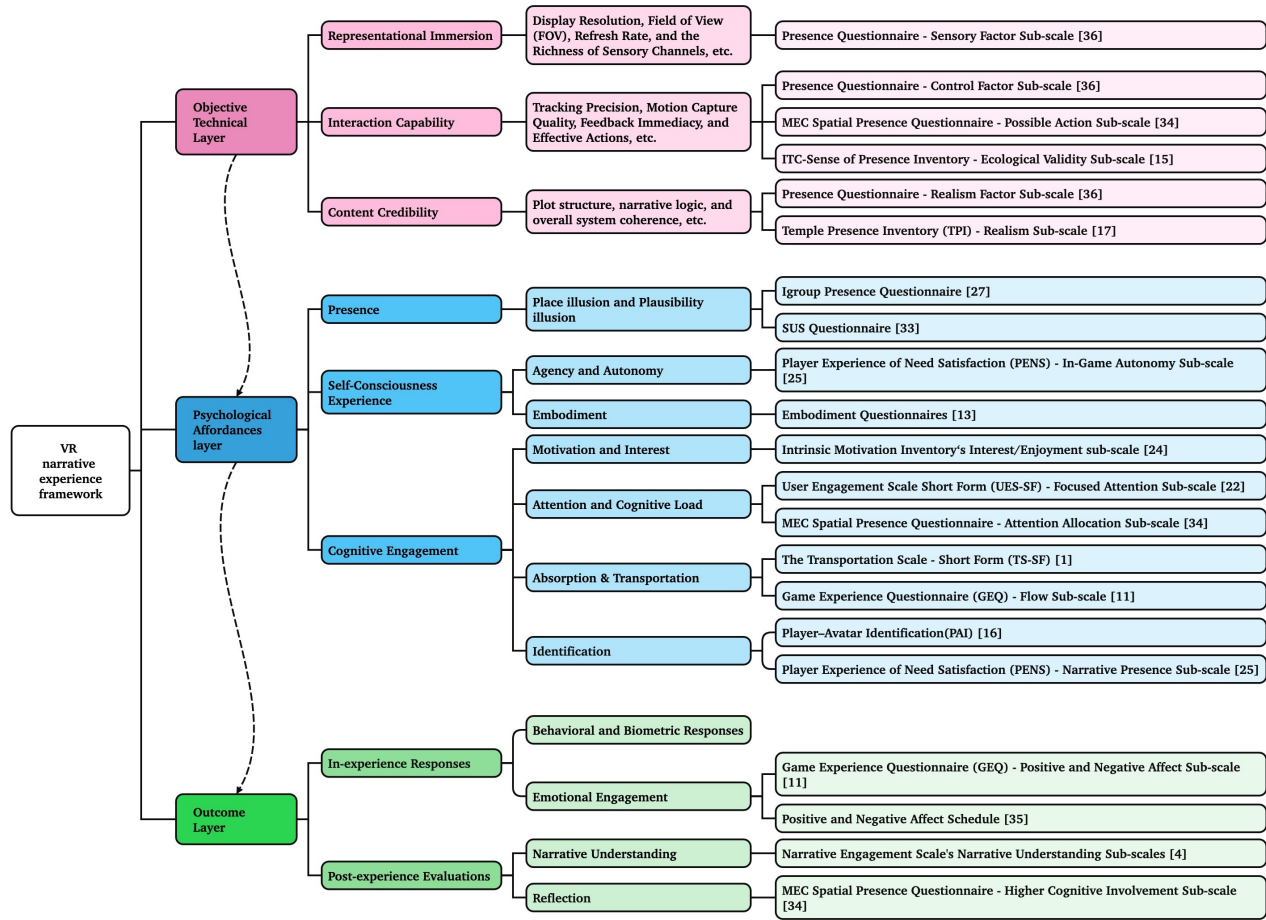


Figure 2: The three-layer classification of VR narrative experiments, detailing internal dimensions and their corresponding measurement scales.

instruments such as the Presence Questionnaire (PQ) [36] and the Temple Presence Inventory (TPI) [17].

Psychological Affordances Layer (Level 2) functions as the "core switch" of the VR narrative experience, bridging objective stimuli to subjective responses; without establishing these psychological illusions, higher-level emotional and physiological outcomes will not occur [18, 36]. This layer is anchored by *Presence*, where the interplay of spatial and plausibility illusions provides a logically consistent stage that prioritizes internal coherence over mere physical realism [28]. Within this stage, *Agency* and *Autonomy* address the user's need for volition and functional capacity [6, 23], while *Embodiment* serves as a powerful mechanism for "identification-as-simulation", allowing users to treat virtual avatars as extensions of themselves [13, 30]. The depth of this experience is further modulated by *Cognitive Engagement*—driven by intrinsic interest and curiosity—and the strategic management of *Attention* and *Cognitive Load*, which ensures mental resources are focused on story progression rather than interface complexities [6, 18, 19]. Collectively,

these affordances facilitate *Absorption* and *Transportation*, fostering a state of intense concentration that enables deep Identification with characters' values and perspectives [14, 32]. As detailed in Fig. 2, these constructs are mapped to validated instruments like the PENS, GEQ, and Transportation Scales [1, 11, 25].

Outcome Layer (Level 3) encompasses the observable consequences of the VR narrative experience at the individual level, serving as a critical external indicator of whether the preceding psychological illusions were successfully established [28]. During the experience, users exhibit *In-experience Responses*, such as physiological arousal and behavioral reactions that mirror real-world responses to virtual threats or events [26, 28]. Beyond biometric data, high-level narrative experiences foster intense *Emotional Engagement*—including empathy and aesthetic appreciation—which transitions the user from mere sensory entertainment to deep emotional investment and reflection [34]. *Post-experience Evaluations* focus on the user's *Narrative Understanding* and the construction of "Situation Models", representing the mental internalizations of

story logic, causality, and character motivations [4, 12]. Ultimately, the framework recognizes *Reflection* as a critical outcome, where narrative transportation successfully alters a user's real-world perspectives or behavioral intentions through emotional persuasion.

Despite the foundational support provided by existing instruments, our analysis identifies three critical bottlenecks in current VR narrative evaluation. First, an over-reliance on hardware parameters leads to a significant dimension gap at the technical level, where "Content Credibility" and the alignment between interaction and narrative logic are often neglected, hindering the accurate assessment of the Plausibility Illusion. Second, a dimensional mismatch arises when applying traditional narrative scales to VR; because immersive media provides direct sensory perception, traditional items measuring "mental imagery" become redundant, yet current tools simultaneously fail to capture VR-specific attributes like Sensorimotor Contingencies and sensory fidelity.

Furthermore, existing frameworks exhibit a functional gap in "Narrative Competence". While established scales like PENS [25] measure general competence, they fail to evaluate whether users feel their actions meaningfully influence the story's moral or emotional trajectory. This omission leaves the user's "narrative efficacy" and long-term aesthetic reflection largely unquantified, highlighting a disconnect between technical immersion and deep narrative engagement. In this case, we introduce 'Narrative Efficacy' to fill the gap left by general scales like PENS. While PENS measures general competence, Narrative Efficacy specifically evaluates the user's perceived agency in altering the story's moral or causal trajectory, providing a more granular metric for interactive storytelling.

Unlike previous descriptive models, this framework also functions as a diagnostic roadmap for VR creators. For instance, if user testing reveals low Narrative Understanding, designers can 'troubleshoot' the experience by tracing back the causal chain: is it caused by excessive Cognitive Load (L2) or a breach in Content Credibility (L1)?

4 Future Research and Conclusion

This study introduces an L1-L2-L3 integrated framework for VR narrative research, synthesizing current innovations to address evaluation tool limitations. By mapping these dimensions, we identify critical measurement gaps and provide a roadmap for experiences with greater plausibility and narrative involvement, moving beyond mere place illusion.

Future research must resolve the "dimensional mismatch" in measurement by developing VR-specific narrative scales. This requires replacing obsolete "mental imagery" metrics with those capturing sensory experiences driven by sensorimotor contingencies (SCs). Finally, balancing the tension between agency and narrative coherence while quantifying "narrative efficacy" will be vital for refining VR evaluation and fostering deeper aesthetic reflection.

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